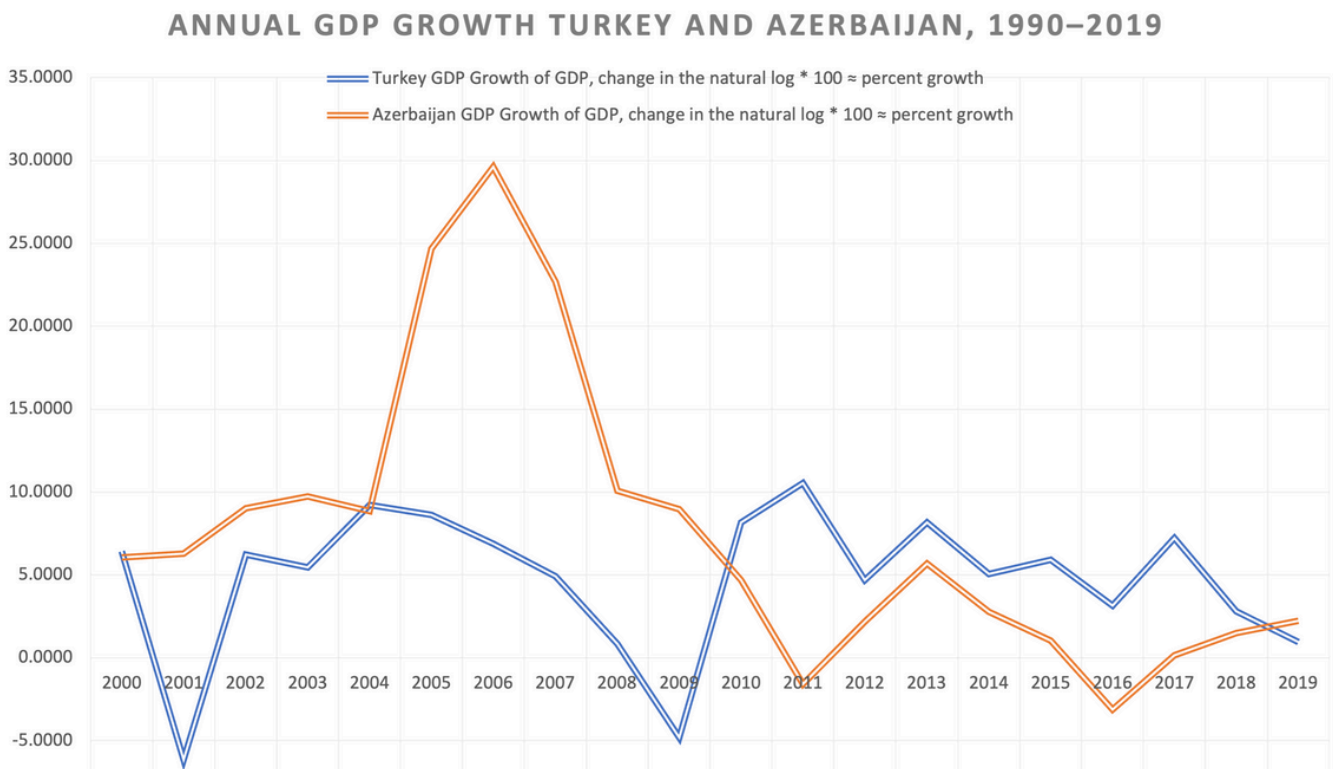
Growth Decomposition: Azerbaijan vs. Turkey

Asian Development Bank
Internal Memorandum

To: Vice President
From: Sally Tsai
Date: Oct. 3rd, 2025
Subject: Comparative Growth Experiences of Asian and European Economies (Azerbaijan, Turkey)

Summary

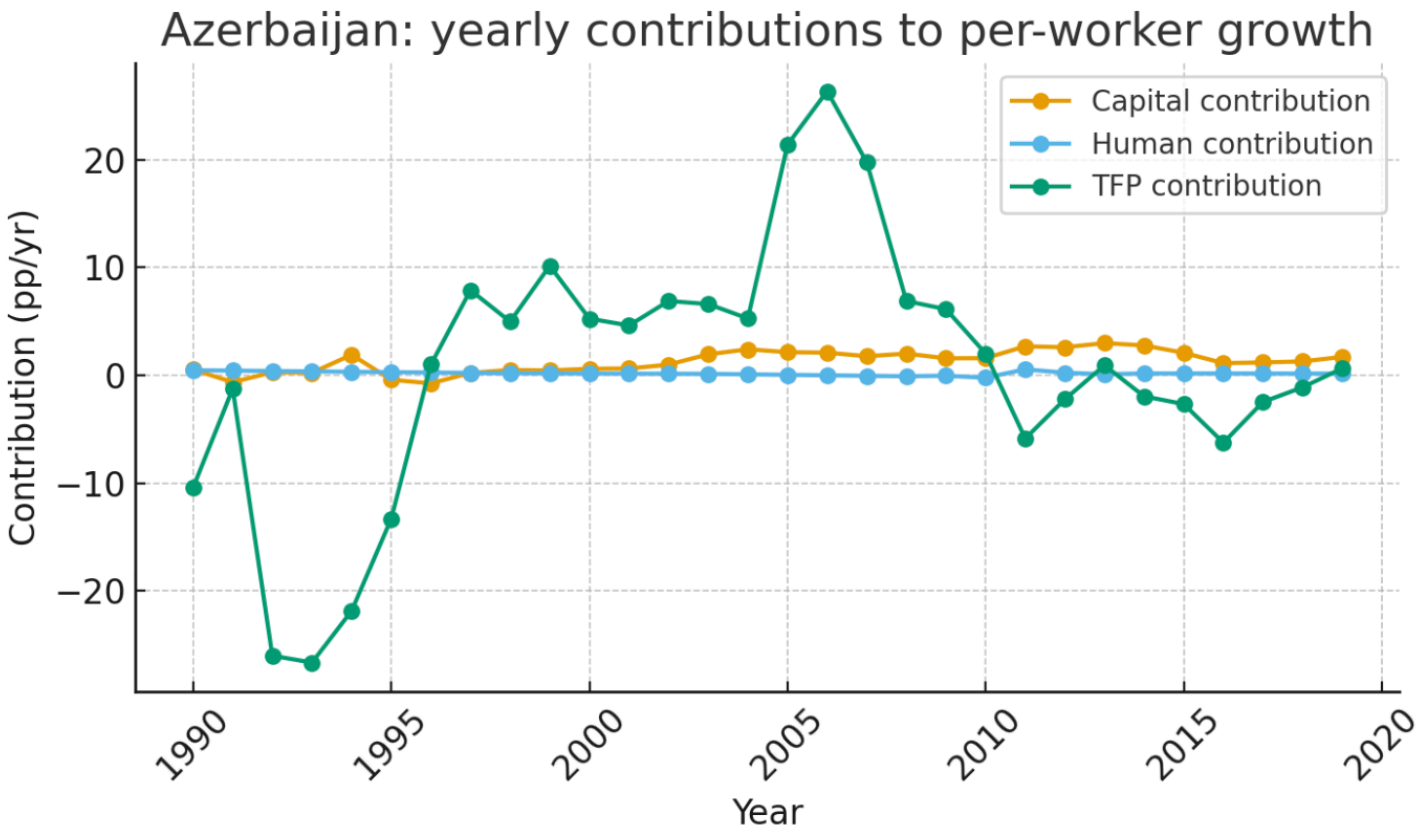


Between 2000 and 2019, Azerbaijan experienced rapid gains in GDP per worker, averaging approximately 6.4 percentage points per year. When we allocate that growth to proximate sources, approximately 70 percent of the increase is attributed to a TFP residual. Measured capital and human capital contributions are relatively small compared to the residual. The timing matters. The large productivity gains occurred mainly in the mid-2000s. That timing aligns with large-scale resource projects and FDI in the country. Taken together, the proximate numbers suggest an ultimate cause linked to resource-led reallocation or concentrated institutional change in a few sectors, rather than broad-based labor or capital accumulation.

Turkey's record is different. Per-worker growth averages about 3.1 percentage points per year. Capital deepening supplies roughly 1.8 percentage points of that total annually, which is roughly 57 percent of per-worker growth. Human-capital improvements add roughly 0.75 percentage points. Measured TFP contributes only about 0.6 percentage points. The proximate pattern is consistent with an investment-driven expansion in which large flows into capital services raise output per worker, but do not translate into significant measured productivity gains. That pattern raises questions about investment quality, sectoral composition, and whether public policy favored non-export or credit-intensive sectors.

Growth Decomposition

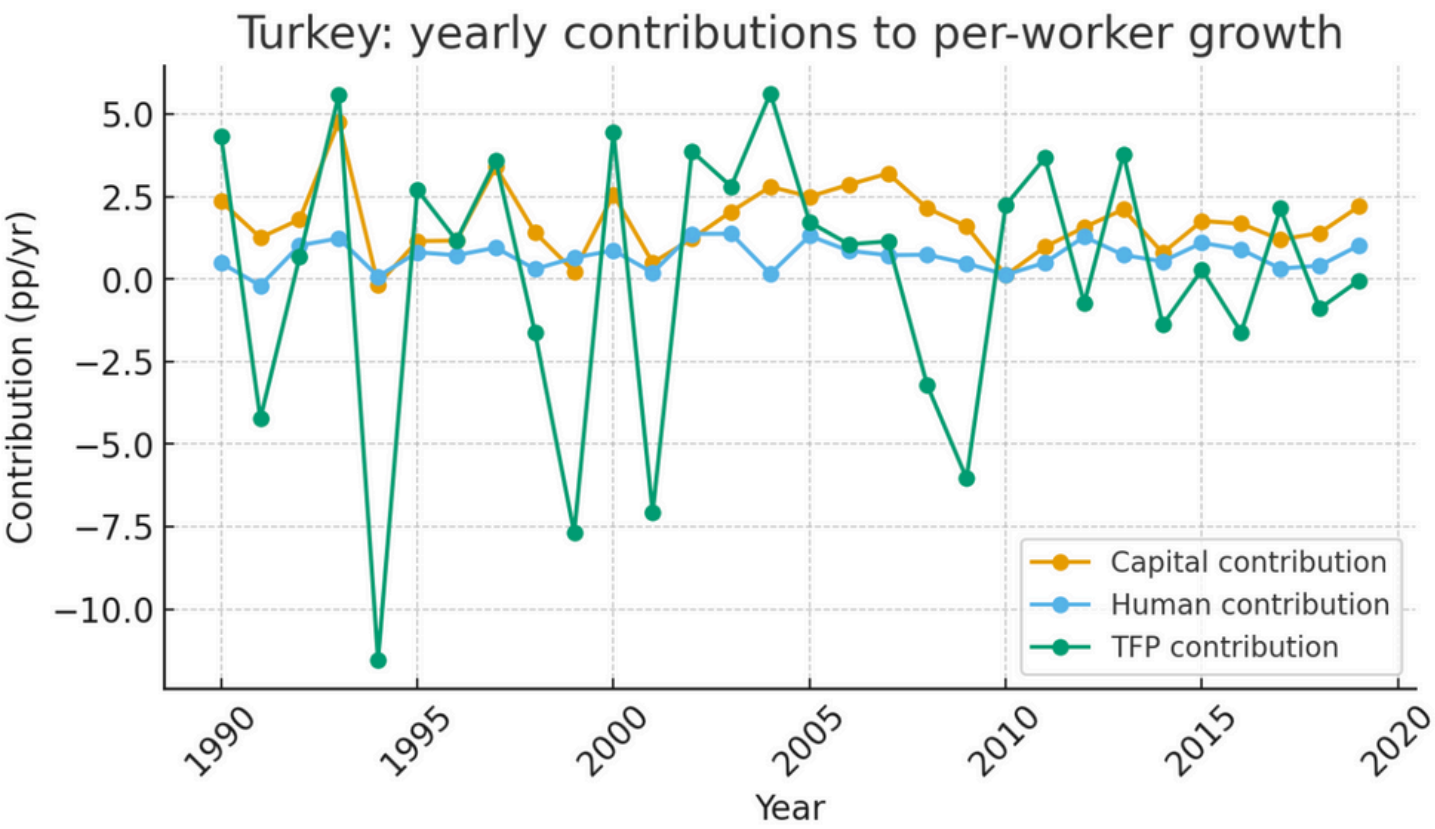
- **Azerbaijan**



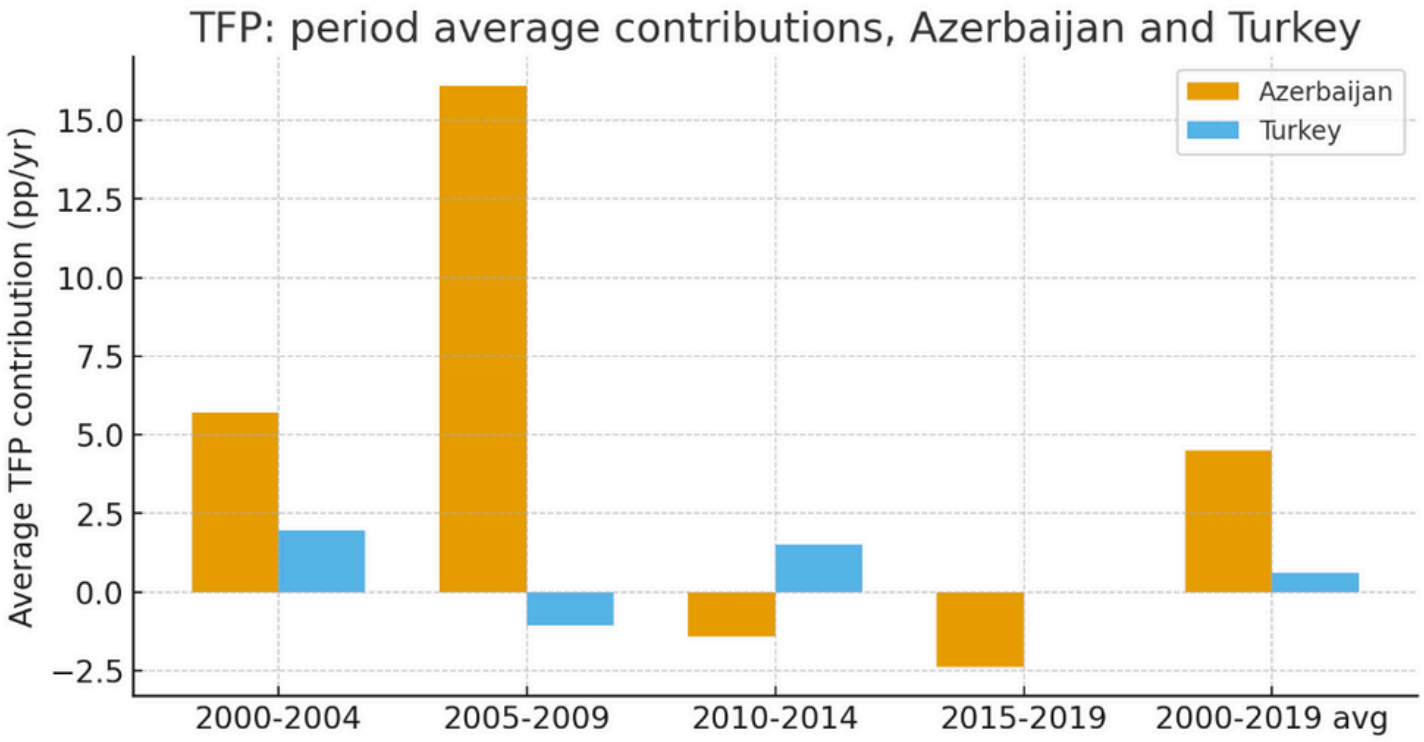
  - **Per-worker Growth (y)**: 6.40% per year. This is the observed increase in output per unit of labor after adjusting for changes in labor quantity. Numerically, this is very large for a two-decade average.
  - **Capital Growth (k)**: about 5.41 pp/yr in capital per worker. Capital accumulation accounts for a moderate share of per-worker growth in levels, but it is significantly smaller than the residual.
  - **Human Capital (h)**: labor-quality growth averages about 1.13% per year, which translates to a human-capital contribution of **≈ 0.75 pp/yr**. Human capital improvements are a non-trivial part of Turkey's story.
  - **TFP residual**: **≈ 4.50 pp/yr**, about 70.3 percent of the total. It means that after accounting for measured factor accumulation, we still see a significant per-worker growth, which we label as productivity. In economic terms, this residual may capture true gains in technology or organization, structural reallocation from low-productivity to high-productivity activities,

capacity utilization effects, or measurement effects due to large resource output or services linked to resource production.

- **Turkey**



  - **Per-worker Growth (y)**: 3.12 pp/yr on average. That is lower than Azerbaijan but still positive.
  - **Capital Growth (k)**: about 5.30 pp/yr in capital per worker, giving a capital contribution of **≈ 1.80 pp/yr**. This is similar in size to Azerbaijan's capital contribution. Turkey's capital deepening is persistent across years rather than concentrated in a short subperiod.
  - **Human capital (h)**: labor-quality growth averages about 1.13% per year, which translates to a human-capital contribution of **≈ 0.75 pp/yr**. Human capital improvements are a nontrivial part of Turkey's story.
  - **TFP residual**: **≈ 0.60 pp/yr**, about 19 percent of per-worker growth. The small residual implies that measured efficiency gains are limited relative to factor accumulation.
- **Nuance**
  - **Azerbaijan:** The mid-2000s TFP spike is concentrated, suggesting that it likely reflects resource-led projects or accounting effects from large fields or pipelines, rather than broad, steady institutional gains. Sectoral or regional tests linking TFP to FDI, oil revenue, or project start dates can help distinguish these mechanisms.



- **Turkey:** Persistent capital deepening, combined with steady labor-quality gains, suggests accumulation rather than economy-wide productivity breakthroughs. We can evaluate sectoral investment shares and credit growth to see whether investment quality or allocation explains the weak TFP response.

Ultimate-cause hypotheses

- ## Azerbaijan:
  - **Hypothesis 1: Resource-led productivity surge (oil, pipeline projects, and FDI)**
    - Large oil and pipeline projects in the 2000s attracted FDI and created supply chain effects. These projects raised measured TFP through rapid capital reallocation, capacity utilization, and firm-level productivity gains in energy-linked sectors.
  - **Method:**
    - **Event study around pipeline/field start**
      - Compare TFP changes in regions/sectors with more direct exposure to the project to those with less exposure. Expect exposed areas to show larger TFP gains after the event if the resource-led reallocation hypothesis holds.
  - **Hypothesis 2: Rapid managerial and institutional shifts concentrated in a few sectors**
    - Post-Soviet reforms or specific privatizations improved management or firm governance in select sectors, resulting in significant measured TFP gains without uniform structural changes.
  - **Method:**
    - **IV with oil-price exposure**
      - Regress TFP on FDI and oil-revenue measures instrumented by world oil price interacted with a pre-2000 oil-exposure index. Expect a positive IV coefficient on FDI if

resource flows caused productivity gains.

- **Turkey:**
  - **Hypothesis 1: Investment and credit expansion fueled capital deepening without strong productivity gains**
    - Recurrent credit expansions and high rates of gross fixed capital formation led to the rapid accumulation of capital services. Much of this investment flowed into credit-sensitive activities where measured returns and productivity gains were limited.
  - **Method:**
    - **Credit-supply IV**
      - Use foreign funding or banking-sector shocks as instruments for domestic credit growth; then test whether exogenous credit explains subsequent capital-deepening and TFP. A null effect on TFP suggests quantity not quality.
  - **Hypothesis 2: Policy-led promotion of non-export, low productivity sectors (construction and protected industries)**
    - Industrial policy and public procurement favored construction and domestically oriented sectors. This raised measured capital stocks, but the investment did not translate into broad efficiency gains, producing a capital-rich but productivity-light growth pattern.
  - **Method:**
    - **Sectoral correlation and DID**
      - Compare TFP trends in tradable manufacturing versus construction and other nontradables around big public works or credit booms. If investment quality is weak, construction will show rising capital but little TFP.

Conclusion

Azerbaijan grew about **6.4 pp/yr** per worker from 2000–2019, but roughly **70%** of that (≈ **4.5 pp/yr**) is a TFP residual concentrated in the mid-2000s, consistent with a resource- or FDI-driven episode rather than broad-based accumulation. Turkey grew about **3.1 pp/yr**, driven mainly by persistent capital deepening (≈ **1.8 pp/yr**) and human-capital gains (≈ **0.75 pp/yr**), with only a small TFP contribution (≈ **0.6 pp/yr**), suggesting accumulation more than economy-wide productivity improvement.